Inference Basics

ECON 490

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Inference in R

In our regression review notes, we considered the following regression:

$$Household\ Income = \beta_0 + \beta_1 Age + u$$

Estimating this using our ACS data in R gives us the output on the right

```
lm(formula = hhincome ~ age, data = graph.data)

Residuals:
    Min      1Q  Median      3Q     Max
-141229  -68518  -21386   51575  268930

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    10735      36505    0.29  0.76912
age             3792        989    3.83  0.00019 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9280 on 148 degrees of freedom
Multiple R-squared:  0.0903,    Adjusted R-squared:  0.0842
F-statistic: 14.7 on 1 and 148 DF,  p-value: 0.000186
```

In these slides, we want to focus on this output. How can we characterize uncertainty in our estimates?

Inference Overview

American Community Survey (ACS) collects data from a ~1% sample of the US

- We can use regression to estimate relationship between age and income
- ***Question:*** What happens if the sample changes?

Inference is how we describe uncertainty in statistical analysis

- ***Standard Errors:*** Measure how much an estimate would vary across samples
- ***P-Values:*** Assess evidence against a hypothesized value

Standard Errors

Goal in any regression analysis is to distinguish signal from noise

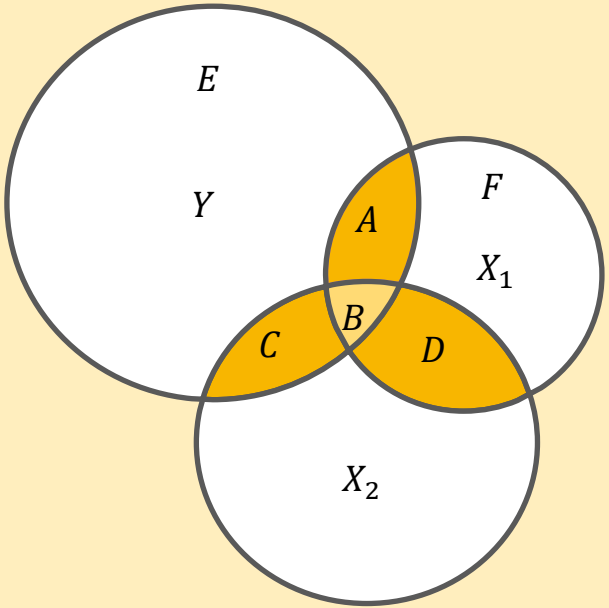
- Assuming no code errors, `lm()` will always give us coefficient estimates
- ***Question:*** How do we know when to take those estimates seriously?

If we drew repeated samples of the ACS, we'd get different estimates of β_1

- The standard error (SE) tells us how much our β_1 estimate would vary
- If our β_1 estimate moves around a lot (i.e., is noisy), it's not very informative

Smaller SE = less sampling variation and a more ***precise*** estimate

Visualizing Precision



E: Unexplained variation in Y

- If E is large, there's a lot of noise in Y
- Reduces the precision of our estimate of β_1

$A + F$: Unique variation in X_1

- If $A + F$ is large, this means X_1 varies a lot independently of other Xs
- More unique variation = more precision

Assume Y is the outcome, X_1 is the explanatory variable we care about, and X_2 is a control variable

Control Variables and Precision

Control variables = explanatory variables that aren't primary focus of your analysis

From last slide, adding X_2 as a control variable could raise or lower SE for $\widehat{\beta_1}$:

- Less residual variation in Y (smaller E): this makes SE for $\widehat{\beta_1}$ ***smaller***
- Less residual variation in X_1 (smaller $A + F$): this makes SE for $\widehat{\beta_1}$ ***larger***

KEY POINT: If omitting X_2 would cause OVB, then we should include it in our model!

- In other words, don't drop a variable just because it affects the precision of the estimate for your primary explanatory variable

P-Values

Statistics like t-stats and p-values are calculated using your $\widehat{\beta_1}$ estimate and SE

In practice, p-values are the easiest way to check statistical significance

- Generally, estimates with p-values less than 0.05 are statistically significant
- What does this mean?

Imagine collecting our data again and estimating the same regression

- What's the probability we observe a t-stat (in absolute value) at least as large as our original value, *if there was actually no "true" effect*?

P-Values using our ACS Regression

General null hypothesis for significance testing: age coefficient is equal to 0

- From our R output, $\widehat{\beta_1} = 3{,}792$, $SE = 989$
- The t-statistic R reports is $t = 3{,}792 \,/\, 989 \approx 3.83$

Under our stated null, $p = 0.00019$

- Tells us the probability of observing a $|t|$ value at least as large in a two-sided test, *assuming the true age coefficient is 0*

A t-stat this large is unlikely if the true effect were actually 0

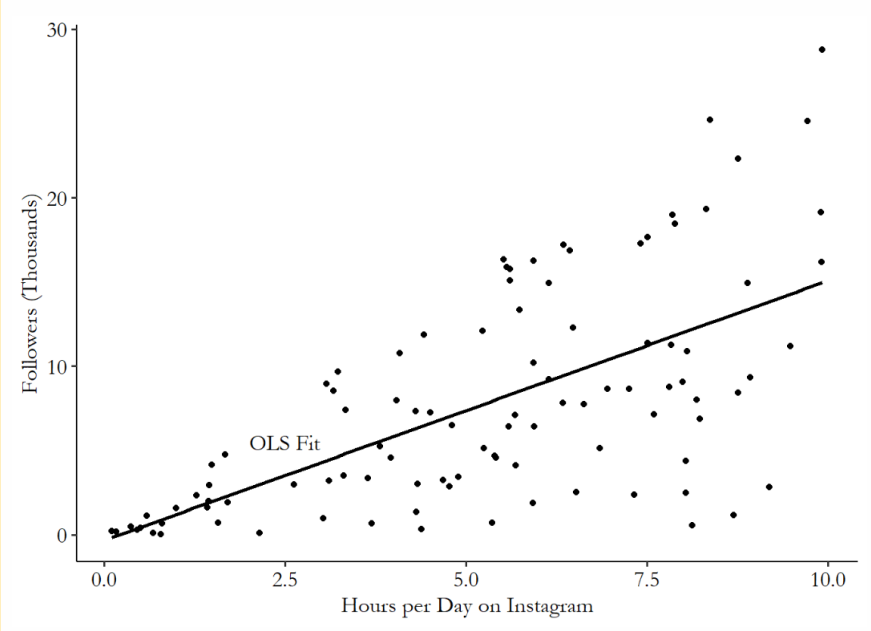
Heteroskedasticity (HK)

HK occurs when our model does a better job explaining Y for some values of X than other values

HK causes our normal SEs to be wrong

In practice, this happens all the time

- As a result, we want a method of accounting for HK
- Do this by using HK-robust SEs



Inference in R

To get HK-robust SEs, use `lm_robust()` function from the `estimatr` package

Two benefits to `lm_robust()`:

- HK-robust SEs + p-values + etc.
- Summary output is easier to work with in R

If you have panel data (e.g., each row is a state/county over a given time period):

- You'll probably run models using geographic unit + time FEs
- If you do so, you should ***cluster*** your SEs on geographic unit via `lm_robust()`

Practical Advice (Pt. 1)

General rule of thumb: p-value less than 0.05 is statistically significant

Practical significance is what ultimately matters

- How large is your estimated effect?
- One approach: try dividing estimated coefficient by mean value of Y

For your capstone analysis, use robust SEs via `lm_robust()` in R

Practical Advice (Pt. 2)

What “breaks” normal inference? What causes problems for regression?

- Main pitfall = having too few observations
- Context matters – how many observations is enough?

A ***very*** rough rule of thumb = you should have ***at least ~50 observations***

- What comparisons do you care about?
- If you care about estimating a difference between groups, then ideally, you’d like ***each*** group to have 50 observations

Sometimes, just having lots of data isn’t enough

- Suppose you have GDP data for the US → lots of years… but only one country
- Solution? Go get data on other countries